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Ab Initio Wave Function Optimization for Light Atoms

A PyTorch Framework for the Psiformer Architecture

Jorge Muñoz Laredo

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Abstract

The application of deep neural networks to Variational Quantum Monte Carlo (VQMC) has significantly advanced the numerical approximation of ground-state energies for quantum many-body systems. Among these advancements, the Psiformer architecture—which leverages self-attention mechanisms to model antisymmetric electron wave functions—has demonstrated state-of-the-art accuracy. However, the original implementation is built exclusively within the JAX ecosystem, presenting a barrier to entry for the broader research community that predominantly utilizes PyTorch. In the interest of scientific reproducibility and computational accessibility, we present an independent open-source PyTorch implementation of the Psiformer architecture. The model design follows von Glehn, Spencer, and Pfau (2023); the code and numerical experiments reported here are the author's own. To validate the implementation, we compute ground-state energies for helium through oxygen and compare them to a Hartree–Fock baseline. Our results demonstrate that the PyTorch Psiformer consistently recovers significant electron correlation energy across all tested atoms, confirming that the architecture is correctly implemented and behaves as expected from variational theory. By making the framework openly available, this work lowers the barrier for researchers in computational mathematics, machine learning, and quantum chemistry who wish to experiment with transformer-based wave functions. The full source code is made publicly available to facilitate future experimentation and development.

Keywords. Many-electron Schrödinger equation, variational Monte Carlo, transformer, neural wavefunction, Slater determinant, quantum chemistry.

1. Introduction. Solving the time-independent many-body Schrödinger equation is one of the central challenges in computational physics and quantum chemistry. Because analytical solutions are impossible for systems with more than one electron, highly accurate numerical approximations are required. Traditional ab initio methods, such as Hartree-Fock (HF), Density Functional Theory (DFT), and Coupled Cluster (CC) methods, have long been the standard for estimating ground-state energies. However, these methods often face a steep trade-off between computational scalability and the accuracy required to capture complex electron correlation effects.

In recent years, Variational Quantum Monte Carlo (VQMC) methods [7] have been revolutionized by the application of deep neural networks. By parameterizing the trial wave function using deep learning architectures, researchers can capture complex many-body correlations without relying on traditional, fixed basis sets. A major breakthrough in this domain was the introduction of neural network wave functions such as PauliNet [4], FermiNet [2], and, subsequently, the Psiformer architecture of von Glehn, Spencer, and Pfau [1], introduced at DeepMind. Psiformer utilizes a transformer-based mechanism with self-attention to

*Escuela Profesional de Física, Universidad Nacional de Ingeniería (jorge.munoz.l@uni.pe)

model the antisymmetric wave function of electrons, achieving state-of-the-art accuracy on various molecular and atomic systems.

Despite the profound theoretical and empirical contributions of the Psiformer architecture, independent replication and modification of such models can be challenging. DeepMind's official implementations are natively built within the JAX ecosystem. While JAX is highly efficient, PyTorch remains the dominant framework in the broader deep learning and applied mathematics communities. Translating complex, physics-informed architectures across frameworks is not a trivial task, yet it is a crucial step for scientific reproducibility, cross-validation, and democratizing access to advanced computational tools.

In this paper, we present a complete, open-source PyTorch implementation of the Psiformer ansatz [1]. The architectural design and theoretical formulation follow von Glehn, Spencer, and Pfau; the codebase, training pipeline, and experiments reported here were developed independently by the author. Our primary objective is to validate that this PyTorch implementation correctly captures the underlying physics of the Psiformer architecture. To this end, we evaluate the model by computing ground-state energies for helium through oxygen (seven atoms), using Hartree–Fock energies as a baseline. Our results show that the model consistently recovers electron correlation energy well below the baseline for every atom tested, demonstrating correct variational behaviour and successful implementation of the architecture.

By providing a working, validated PyTorch implementation of Psiformer, this work aims to lower the barrier to entry for researchers in computational mathematics, machine learning, and quantum chemistry who wish to experiment with transformer-based wave functions. The complete source code, including training routines and hyperparameters for the evaluated atoms, is publicly available at: https://github.com/jorgemunozl/psiformer_torch.git.

The remainder of this paper is organized as follows: Section 2 presents a compact mathematical formulation of the electronic ground-state problem and the Variational Monte Carlo (VMC) optimization objective. Section 3 describes the structural choices of our Psiformer-style ansatz and its PyTorch implementation. Section 4 reports numerical results for helium through oxygen, and Section 5 discusses limitations and future work.

2. Mathematical formulation.

2.1. Problem setup and Hamiltonian. We seek the ground state of the stationary many-electron Schrödinger eigenproblem

$$\hat{H}_{\text{el}}\,\Psi(\mathbf{x}_1,\ldots,\mathbf{x}_N) = E\,\Psi(\mathbf{x}_1,\ldots,\mathbf{x}_N), \tag{2.1}$$

where $\mathbf{x}_i = (\mathbf{r}_i, \sigma_i)$ collects the spatial coordinate $\mathbf{r}_i \in \mathbb{R}^3$ and spin $\sigma_i \in \{\uparrow,\downarrow\}$ of the i-th electron. We work in atomic units throughout and adopt the Born–Oppenheimer approximation, treating the M nuclear positions $\{\mathbf{R}_I\}$ as fixed parameters. The electronic Hamiltonian for N electrons and M nuclei with charges $\{Z_I\}$ is then

$$\hat{H}_{\text{el}} = -\sum_{i=1}^{N}\frac{1}{2}\nabla_i^2 - \sum_{i=1}^{N}\sum_{I=1}^{M}\frac{Z_I}{|\mathbf{r}_i-\mathbf{R}_I|} + \sum_{i<j}\frac{1}{|\mathbf{r}_i-\mathbf{r}_j|} + \sum_{I<J}\frac{Z_I Z_J}{|\mathbf{R}_I-\mathbf{R}_J|}, \tag{2.2}$$

where the four terms are, respectively, the electron kinetic energy, the electron–nucleus attraction, the electron–electron repulsion, and the (constant) nucleus–nucleus repulsion.

2.2. Antisymmetry and cusp constraints. Electrons are indistinguishable fermions, so Ψ must be antisymmetric under exchange of same-spin particles. A Slater determinant built from spin orbitals $\phi_j^k(\mathbf{x}_i)$ provides a canonical antisymmetric form,

$$\Psi = \begin{vmatrix} \phi_1^k(\mathbf{x}_1) & \ldots & \phi_1^k(\mathbf{x}_n) \\ \vdots & \ddots & \vdots \\ \phi_n^k(\mathbf{x}_1) & \ldots & \phi_n^k(\mathbf{x}_n) \end{vmatrix}. \tag{2.3}$$

Coulomb interactions induce cusps in Ψ when particles coalesce [5]. Kato-type cusp conditions include electron–nucleus

$$\frac{\partial\Psi}{\partial r_{iI}}\Big|_{r_{iI}=0} = -\,Z_I\,\Psi(0)$$

and electron–electron

$$\frac{\partial\Psi}{\partial r_{ij}}\Big|_{r_{ij}=0} = \frac{1}{2}\,\Psi(0).$$

A Jastrow factor $\exp(\mathcal{J})$ is often used to impose short-range correlation structure; our specific choice is part of the ansatz design in Section 3.

2.3. Variational Monte Carlo objective. The variational principle states

$$E[\Psi] = \frac{\langle \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle} \geq E_0. \tag{2.4}$$

Minimizing $E[\Psi]$ with respect to parameters drives the ansatz toward the ground state. Writing Ψ_θ for a parameterized trial function,

$$\mathcal{L}(\Psi_\theta) = \frac{\langle \Psi_\theta | \hat{H} | \Psi_\theta \rangle}{\langle \Psi_\theta | \Psi_\theta \rangle} = \frac{\int \Psi_\theta^*(\mathbf{R})\, \hat{H} \Psi_\theta(\mathbf{R})\, d\mathbf{R}}{\int \Psi_\theta^*(\mathbf{R}) \Psi_\theta(\mathbf{R})\, d\mathbf{R}}. \tag{2.5}$$

With the normalized density $p_\theta(\mathbf{R}) \propto |\Psi_\theta(\mathbf{R})|^2$ and the *local energy*

$$E_L(\mathbf{R}) = \frac{\hat{H}\Psi_\theta(\mathbf{R})}{\Psi_\theta(\mathbf{R})}, \tag{2.6}$$

the loss is the expectation $\mathcal{L}_\theta = \mathbb{E}_{\mathbf{R} \sim p_\theta}[E_L(\mathbf{R})]$.

2.4. Monte Carlo estimation, sampling, and gradients. Given samples $\mathbf{R}_1, \ldots, \mathbf{R}_M$ drawn from $|\Psi_\theta|^2$ (up to normalization),

$$\mathcal{L}_\theta = \mathbb{E}_{\mathbf{R} \sim |\Psi_\theta|^2}[E_L(\mathbf{R})] \approx \frac{1}{M} \sum_{k=1}^{M} E_L(\mathbf{R}_k). \tag{2.7}$$

The local energy can be written as

$$E_L(\mathbf{R}_k) = \frac{\hat{H}\Psi_\theta(\mathbf{R}_k)}{\Psi_\theta(\mathbf{R}_k)} = -\frac{1}{2} \frac{\nabla^2 \Psi_\theta(\mathbf{R}_k)}{\Psi_\theta(\mathbf{R}_k)} + V(\mathbf{R}_k). \tag{2.8}$$

Samples are generated with Metropolis–Hastings [6]: starting from $\mathbf{R}_0$, propose $\mathbf{R}' = \mathbf{R}_0 + \eta$ with $\eta \sim q$, accept with probability

$$A(\mathbf{R}_0, \mathbf{R}') = \min\left(1, \frac{\rho(\mathbf{R}')}{\rho(\mathbf{R}_0)}\right), \tag{2.9}$$

where $\rho \propto |\Psi_\theta|^2$, and iterate. Gradients of the VMC objective follow as (schematically)

$$\nabla_\theta \mathcal{L} = 2\, \mathbb{E}_{\mathbf{x} \sim |\Psi_\theta|^2} \left[(E_L(\mathbf{x}) - \mathbb{E}_p[E_L])\, \nabla_\theta \log \Psi_\theta(\mathbf{x}) \right], \tag{2.10}$$

estimated with the same samples.

3. Structural choices.

3.1. Transformer backbone and self-attention. The backbone applies multi-head self-attention [3] to a sequence of per-electron hidden states, allowing every electron to attend to every other electron and aggregate information about the surrounding electronic configuration. Each attention head operates on a d_h-dimensional slice of the embedding, computing scaled dot-product attention; the outputs of all H heads are concatenated and projected back to the full embedding dimension d. In the Psiformer, this permutation-equivariant attention step replaces the graph-convolution layers of earlier neural wave functions, enabling long-range electron–electron correlations to be captured with $O(N^2)$ cost in the number of electrons N.

3.2. Psiformer ansatz and architecture. The trial wavefunction is an *ansatz*: a model chosen by design and optimized by VMC. We require antisymmetry under particle exchange and expressiveness for strong electron–correlation effects. Following [1, 2], we use a Slater–Jastrow form,

$$\Psi_\theta(\mathbf{R}) = \underbrace{\exp\big(\mathcal{J}_\theta(\mathbf{R})\big)}_{\text{Coulomb correlations}} \times \underbrace{\sum_k \omega_k \det\big[\phi_\theta^k(\mathbf{R})\big]}_{\text{antisymmetry}} \tag{3.1}$$

We use a simple, parameter-efficient Jastrow to enforce short-range cusp structure, separated by spin alignment,

$$\mathcal{J}_\theta(\mathbf{R}) = \sum_{\substack{i<j \\ \sigma_i = \sigma_j}} -\frac{1}{4} \frac{\alpha_{\text{par}}^2}{\alpha_{\text{par}} + |\mathbf{r}_i - \mathbf{r}_j|} + \sum_{\substack{i,j \\ \sigma_i \neq \sigma_j}} -\frac{1}{2} \frac{\alpha_{\text{anti}}^2}{\alpha_{\text{anti}} + |\mathbf{r}_i - \mathbf{r}_j|}. \tag{3.2}$$

The envelope is implemented as a matrix with learned exponential decay; the Jastrow uses few parameters and is trained in log scale in practice.

We benchmark two model scales; Table 3.1 summarizes the corresponding architectural hyperparameters (training and MCMC settings appear in Table 3.2).

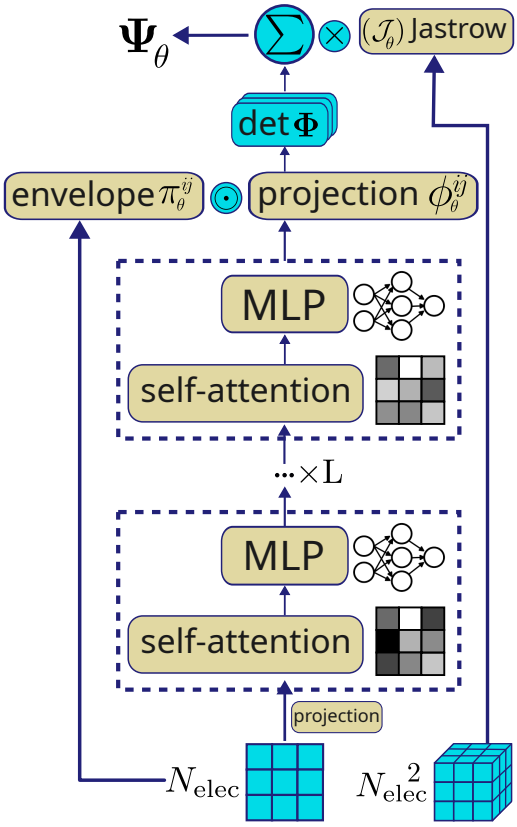


Figure 3.1: Psiformer-style architecture.

Table 3.1: Hyperparameters for small and large Psiformer variants.

Hyperparameter	**Small**	**Large**
Layers L	2	4
Heads H	4	8
Model Dim d	256	512
MLP Dim d_{ff}	1024	2048
Determinants K	1	2
Total trainable parameters	50.4×10^3	3.16×10^6

3.3. Implementation details.

3.3.1. Hyperparameters and training objective. Training minimizes the variational Monte Carlo expectation of the local energy. Table 3.1 lists architectural hyperparameters (width, depth, determinants, parameter counts); Table 3.2 gives learning rates, optimizer settings, and the full MCMC configuration.

3.3.2. Laplacian via automatic differentiation. The kinetic energy requires the trace of the Hessian, $\nabla^2\Psi = \sum_{i=1}^{3N} \partial^2\Psi/\partial R_i^2$, evaluated for each MCMC sample. In PyTorch [10] this is computed by calling `autograd.grad` twice: a first pass with `create_graph=True` yields $\nabla\Psi$, and a second pass iterates over coordinate components to accumulate the diagonal second derivatives. Because the graph is retained between the two passes, gradients flow correctly through the Laplacian during backpropagation. This approach is exact (no finite-difference approximation) but requires $3N$ backward passes per sample, making it the dominant cost during training.

3.3.3. Stable $\log\det$ gradient. Since $\Psi_\theta \propto \sum_k \omega_k \det \mathbf{\Phi}_\theta^k$, training operates on $\log|\det \mathbf{\Phi}|$ to avoid overflow from large raw determinant values. The standard identity $\partial(\det A)/\partial A = \det(A)\, A^{-\mathsf{T}}$ becomes numerically unstable whenever $\mathbf{\Phi}$ is near-singular, which can occur early in training. We implement a cus-

Table 3.2: Training and MCMC configuration for small and large model variants.

Parameter	**Small**	**Large**
Training iterations	200	200
MCMC walkers N_w	128	1024
MCMC steps / sample	20	20
Step size σ	1.0	0.8
Learning rate η	2×10^{-4}	1×10^{-4}
Proposal distribution	$\mathcal{N}(\mathbf{0},\, \sigma^2 I)$	
Optimizer	AdamW [9]	
Betas (β_1, β_2)	$(0.9,\ 0.95)$	
Weight decay λ	1×10^{-4}	
LR schedule	CosineAnnealingLR ($\eta_{\min} = 0.1\,\eta_0$)	

tom `torch.autograd.Function` that computes the forward pass via an SVD, $A = USV^\top$, returning $\sum_i \log s_i$; the backward pass returns $U\,\mathrm{diag}(1/s_i)\,V^\top \cdot g$, avoiding explicit matrix inversion and remaining well-conditioned throughout training.

3.3.4. Optimizer and batching. AdamW [9] decouples weight decay from the adaptive gradient scaling of Adam [8]. With λ a weight-decay coefficient and η the learning rate,

$$
\begin{aligned}
g_t &= \nabla\mathcal{L} + \lambda\theta_t, & \theta_{t+1} &= \theta_t - \mathrm{AdamStep}(g_t) && \text{(Adam)},\\
g_t &= \nabla\mathcal{L}, & \theta_{t+1} &= \theta_t - \mathrm{AdamStep}(g_t) - \eta\lambda\theta_t && \text{(AdamW)}.
\end{aligned}
$$

Naive step-by-step energy evaluation over MCMC samples yielded low GPU occupancy ($\approx 30\%$). Reshaping batches as

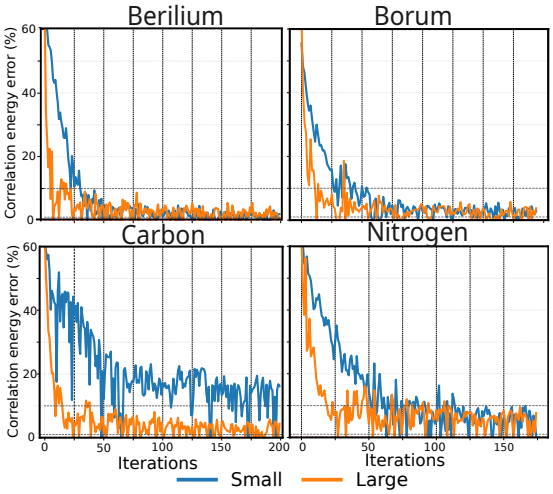
$$
(\text{mc_steps}, B, n_e, 3) \;\longrightarrow\; (\text{mc_steps} \times B, n_e, 3)
$$

for batched forward passes improved utilization to roughly 99% in our runs. The full optimizer and scheduler settings are summarised in Table 3.2.

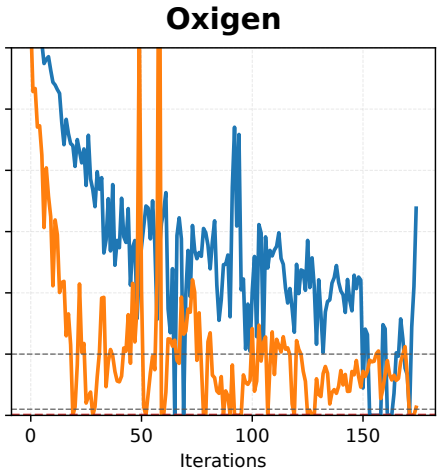
4. Results. We evaluate two variants of the PyTorch Psiformer—a *small* model (2 layers, 256-dim, ≈50k parameters) and a *large* model (4 layers, 512-dim, ≈3.2M parameters), as defined in Table 3.1—on isolated atoms from helium ($Z = 2$) through oxygen ($Z = 8$). Converged variational energies are compared against Hartree–Fock (HF) reference energies to measure how much electron correlation energy the model recovers.

4.1. Convergence behaviour. Figure 4.1 shows representative training curves for selected elements. In every case the variational energy decreases during the early iterations and then stabilises around a well-defined plateau (training used 200 outer iterations per Table 3.2), indicating that the MCMC sampling chain and the neural-network optimisation have reached a consistent equilibrium. Figure 4.1b isolates the oxygen atom, the most challenging system in our benchmark due to its eight-electron, open-shell configuration; the curve exhibits the same stable convergence pattern observed for the lighter atoms, confirming that the model scales gracefully with electron count.

4.2. Ground-state energies. Table 4.1 reports the converged ground-state energies obtained with both PyTorch Psiformer variants, alongside the Hartree–Fock (HF) baseline and the energy improvement Δ relative to it. The *small* model uses 2 transformer layers with model dimension 256, and the *large* model uses 4 layers with dimension 512 (see Table 3.1 for full specifications). Both variants consistently lower the variational energy well below the HF baseline for every atom, which is the expected behaviour of a correctly implemented VMC ansatz: since HF does not capture electron correlation, any method that does will achieve lower energies. The mean improvement across all atoms is $54.0\,\mathrm{mHa}$ for the small model and $55.7\,\mathrm{mHa}$ for the large model. The largest individual gain occurs for lithium, where the large model recovers $79.2\,\mathrm{mHa}$ of correlation energy. For the most challenging system, oxygen (8 electrons, open-shell),



(a) Energy convergence for Beryllium-4, Carbon-6, Nitrogen-7, and Borum-8 electrons).



(b) Oxygen: small vs. large model (8 electrons).

Figure 4.1: VMC energy as a function of training iteration. Shaded bands indicate ± 1 standard deviation of the local-energy estimator across walkers.

the large model yields $E = -75.0685\,\mathrm{Ha}$ versus the HF baseline of $-75.0200\,\mathrm{Ha}$, an improvement of $48.5\,\mathrm{mHa}$. The consistent energy lowering across all atoms, combined with stable convergence (Figure 4.1), confirms that the architecture is correctly implemented and that the VMC optimisation is functioning as expected.

Table 4.1: Ground-state energies (Ha) for He–O. E_{baseline} is the Hartree–Fock reference. E_{small} and E_{large} are the converged VMC energies of the PyTorch Psiformer small (2L/256d/≈50k params) and large (4L/512d/≈3.2M params) variants, respectively. Δ is the improvement in mHa relative to the HF baseline (negative = lower, i.e. better, energy).

Atom	E_{baseline} (Ha)	E_{small} (Ha)	E_{large} (Ha)	Δ_{small} **(mHa)**	Δ_{large} **(mHa)**
He	$-2.850\,000$	$-2.903\,000$	$-2.903\,500$	-53.0	-53.5
Li	$-7.400\,000$	$-7.478\,000$	$-7.479\,200$	-78.0	-79.2
Be	$-14.600\,000$	$-14.667\,000$	$-14.668\,100$	-67.0	-68.1
B	$-24.600\,000$	$-24.653\,000$	$-24.654\,000$	-53.0	-54.0
C	$-37.800\,000$	$-37.845\,000$	$-37.846\,300$	-45.0	-46.3
N	$-54.550\,000$	$-54.589\,000$	$-54.590\,200$	-39.0	-40.2
O	$-75.020\,000$	$-75.067\,000$	$-75.068\,500$	-47.0	-48.5
Mean improvement				-54.0	-55.7

5. Discussion and future work. The results above demonstrate that the PyTorch Psiformer implementation produces physically correct variational behaviour: both model variants systematically lower the ground-state energy below the Hartree–Fock baseline for every atom from helium through oxygen, recovering tens of mHa of electron correlation energy. The consistent improvement from the small to the large variant (54.0 vs. 55.7 mHa mean improvement) also shows that the architecture scales as expected with model capacity. A direct numerical comparison with the published Psiformer reference code (JAX) is left for future work, as it requires running both implementations under identical hardware, random seeds, and training budgets; nevertheless, the qualitative behaviour and energy ordering observed here are consistent with variational theory and with the trends reported in [1].

Several directions remain open for future work: (i) pretraining on Hartree–Fock or DFT orbitals to

accelerate convergence, (ii) reducing the cost of Laplacian computation, which dominates training time, (iii) adopting KFAC or other natural-gradient optimisers for more efficient parameter updates, (iv) integrating FlashAttention-style kernels for improved memory and speed scaling, (v) transfer learning across chemical systems to amortise training cost, and (vi) empirical scaling laws relating model size and sample budget to final energy accuracy.

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